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Assessing the Potential of Free Remote Sensing Data for Automated Agricultural Mapping: A Comparative Evaluation of KNN and Random Forest Algorithms as Tools for Supporting Land Resource Management

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Abstract

Accurate and up-to-date mapping of agricultural crops constitutes a critical infrastructure for land resource management and environmental planning. However, the high dynamism of agricultural areas and the phenological similarity between different crops pose a significant challenge to automated identification based on satellite imagery. This study examines the potential of combining NDVI time series derived from free satellite data (Sentinel-2) with machine learning algorithms for the spatial classification of crops in the Yoav Regional Council. The study was based on a comparative evaluation of a tree-based ensemble model (Random Forest) and a distance-based model (K-Nearest Neighbor).

The findings indicate the clear superiority of the Random Forest model, which achieved a higher overall accuracy (51.2% vs. 46.1%) and produced homogeneous classification maps that faithfully reflect the agronomic reality on the ground, while filtering out "noise" and reducing the pixelation effect. In addition, the analysis shows that both models performed very well (over 71% accuracy for RF) in identifying evergreen orchards (avocado and olive), but struggled to separate seasonal field crops (such as wheat and clover) owing to near-complete phenological overlap. The study concludes that combining free Sentinel-2 data with the RF algorithm constitutes a practical, accessible, and cost-effective tool for environmental and agricultural monitoring. Furthermore, the findings pave the way for focused follow-up studies that will integrate object-based image analysis (OBIA) and complementary structural data (such as SAR radar from the Sentinel-1 satellite and texture analysis) in order to overcome the barriers to spectral separation and improve mapping accuracy.

1. Introduction

Accurate and up-to-date mapping of land use, and in particular of agricultural crop types, constitutes a critical information infrastructure for natural resource management, environmental planning, and ecological risk assessment. While traditional field surveys or manual interpretation provide highly reliable data, these processes are resource-intensive, time-consuming, and difficult to implement or update frequently at the regional scale. Remote Sensing technologies and Geographic Information Systems (GIS) offer an efficient, cost-effective, and continuous technological alternative for monitoring the Earth's surface at a broad scale, while reducing dependence on physical fieldwork.

Unlike static land covers such as built-up areas or infrastructure, agricultural areas are characterized by particularly high dynamism. This dynamism stems from the natural cyclicity of vegetation (phenology) and from deliberate agricultural practices such as crop rotation. Consequently, identifying the Crop Type on the basis of single-date imagery is usually insufficient. Different crops may exhibit an identical Spectral Signature at certain developmental stages, for example when two fields are both at peak green-up or after harvest. Accordingly, the recent research literature points to the essential need for Multi-temporal Analysis. Such analysis tracks changes in vegetation indices, such as the Normalized Difference Vegetation Index (NDVI), which is calculated from the ratio between the NIR and Red bands:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

This continuous monitoring throughout the entire growing season enables the construction of a "phenological-spectral signature" unique to each crop type. As demonstrated in the research literature, the use of dense time series makes it possible to successfully overcome the limitations of single images and provides the temporal resolution required for the accurate identification of growth stages (Griffiths et al., 2019; Veloso et al., 2017).

Studies emphasize that integrating multi-seasonal data is critical for mapping in complex agricultural environments, since it captures phenological variability across the growing seasons and makes it possible to distinguish with high accuracy between crops that exhibit similar spectral signatures within certain time windows (Rodriguez-Galiano et al., 2012; Vuolo et al., 2018).

The availability of data from satellites with high spatial and temporal resolution, such as the Sentinel-2 constellation of the European Space Agency (ESA), has opened new avenues for agricultural monitoring. These sensors provide imagery at a revisit frequency of approximately five days, enabling close tracking of growth stages (Bontemps et al., 2015). However, handling the enormous volumes of information (Big Data) requires a shift from basic classification methods to advanced Machine Learning algorithms. This study focuses on a comparison between two major classification approaches: the Random Forest (RF) algorithm,

which is based on an ensemble of decision trees, and the K-Nearest Neighbors (KNN) algorithm, which is based on similarity and distance in the spectral feature space. The RF algorithm is particularly widely used in mapping based on multi-temporal data, owing to its ability to efficiently process high-dimensional spectral datasets without succumbing to Overfitting (Belgiu & Drăguţ, 2016).

Already in the pioneering studies conducted immediately after the launch of the Sentinel-2 satellite, it was demonstrated that combining its spatial and spectral resolution with the learning model of the RF algorithm offers unprecedented potential for producing reliable and accurate land cover maps in complex agricultural environments (Bontemps et al., 2015; Immitzer et al., 2016). The ability of these models to detect latent patterns within spatial data enables automated pixel classification at a high level of accuracy, based on Ground Truth data used to train the model.

In the Israeli context, and particularly in the Yoav Regional Council, the challenge of automated classification is intensified by the nature of local agriculture: intensive, highly diverse agriculture spread over relatively small land parcels. Developing a reliable classification model in this complex environment constitutes a powerful tool. Beyond the mapping capability itself, building a model capable of identifying crops remotely constitutes a key component in analyzing the potential exposure of the population to specific pesticides, which are directly derived from the type of crop grown in proximity to their place of residence.

Research Objectives

The overarching objective of this study is to develop, implement, and validate a machine learning–based methodology for the automated identification and classification of agricultural crop types in the Yoav Regional Council, using free satellite imagery from the Sentinel-2 constellation.

The specific objectives include:

1. **Comparison of classification performance:** Evaluating the accuracy of the Random Forest algorithm against the KNN algorithm in mapping five selected crop types.
2. **Multi-temporal analysis:** Examining the contribution of NDVI time series to the ability to separate crops with similar spectral signatures over the course of the growing season.

Accuracy Assessment: Validating the spatial results using a Confusion Matrix and the Overall Accuracy and Kappa metrics, based on ground truth data.

2. Methods

2.1 Study Area and Data Sources

The study focuses on the agricultural area of the Yoav Regional Council, a region characterized by intensive, diverse, and frequently changing agricultural activity, which constitutes an ideal test environment for developing classification models. The spatial database for the study relies on the integration of remote sensing data with administrative vector data:

- **Satellite Imagery:** The spatial analysis is based on Time-Series of imagery from the Sentinel-2 satellite of the European Space Agency (ESA). The images were downloaded at an atmospherically corrected processing level (Level 2A – Bottom of Atmosphere) with a spatial resolution of 10 meters. In order to monitor the phenology (growth cycle) of the different crops, images from several key dates throughout the agricultural growing season were selected. Rigorous filtering was performed to retain only images with cloud cover below 10%, in order to prevent masking of the land surface and to minimize errors in spectral reflectance.
- **Ground Truth:** The Target Variable was derived from a vector polygon layer (Shapefile) of the "Public Agricultural Parcels" type maintained by the Ministry of Agriculture. These data underwent preliminary filtering so that only parcels whose "update date" value exactly matched the year of the satellite imagery were included. This process is essential for ensuring research traceability and for preventing statistical biases arising from farmers' crop rotation practices.

2.2 Spatial Pre-processing

The initial processing chain and the preparation of the explanatory variables (Feature Engineering) were carried out in the ArcGIS Pro software environment. First, all satellite images were geometrically clipped to the boundaries of the study area using the Clip tool, in order to streamline the computational processes.

For each monthly time stamp, the Normalized Difference Vegetation Index (NDVI) was calculated; this index serves as a quantitative and stable indicator of the level of growth and photosynthetic activity of the plant. The index is calculated on the basis of the normalized difference between the reflected radiation in the near-infrared (NIR) band and the visible red (Red) band, according to the formula:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

The NDVI layers, together with the standard color bands (RGB) from all sampled months, were merged into a Multi-temporal Raster Stack using the Composite Bands tool. This methodological approach enables the construction of a spectral-temporal signature for each pixel in the study area, representing the development of vegetation over time, rather than relying on an instantaneous and static signature.

2.3 Machine Learning Architecture and Classification Workflow

The classification of land cover and agricultural crops was carried out using a comparative approach, employing two supervised machine learning algorithms: K-Nearest Neighbors (KNN) as a baseline model, and Random Forest (RF) as an advanced Ensemble model. Whereas the KNN algorithm classifies each pixel on the basis of statistical similarity (Euclidean distance) to the K pixels closest to it, the RF model is based on constructing a "forest" of random decision trees and aggregating their predictions. The Random Forest algorithm was selected for this task because it is a robust and efficient model, capable of successfully handling large data volumes and multiple input variables from the multi-temporal NDVI indices without suffering from Overfitting (Griffiths et al., 2019; Vuolo et al., 2018).

The process was implemented in the ArcGIS Pro environment and comprised three consecutive stages:

3. **Training Samples Generation:** Using the Training Samples Manager pane, a hierarchical classification schema was built based on the "Crop Category" and "Crop Name" fields from the Ministry of Agriculture ground truth polygons. From these polygons, representative training areas were sampled for each crop type. Each sample was assigned a category name and a unique numerical value, which aggregated the pixel data from the multi-temporal rasters. The exact same training samples were used to feed both models.
4. **Model Training:** Training was performed separately for each algorithm using the Train K-Nearest Neighbor Classifier and Train Random Forest Classifier tools. At this stage, the structural parameters of the models were defined: for the RF model, 100 decision trees (Number of Trees) were used to ensure statistical stability, and for the KNN model, a value of K = 1 (Number of Neighbors) was set in order to classify on the basis of the single nearest neighbor in the feature space. This setting is consistent with parameter sensitivity studies showing that the predictive accuracy of the RF algorithm tends to stabilize when 100 decision trees are used (Maxwell et al., 2018). The models were fed with the multi-band file combining the temporal data (which constitutes the set of explanatory variables) and with the training polygon layer. Upon completion of data processing, the system produced two trained model definition files (in .ecd format).
5. **Raster Classification:** The trained models were applied to all pixels in the study area using the Classify Raster tool. The final output consisted of two classified rasters in Esri's Cloud Raster Format (CRF), enabling a spatial comparison between the predictions of the KNN model and those of the RF model.

2.4 Accuracy Assessment and Quality Control

In order to validate the significance of the results and to compare the performance of the models objectively, a rigorous quality control framework was established, based on independent data that did not take part in the training stage:

- **Sampling of control points:** Using the Create Accuracy Assessment Points tool, a layer of ground truth points was generated within selected polygons from the Ground Truth data. The sampling strategy was implemented using the Equalized Stratified Random sampling method, with an allocation of 100 points to each of the 5 crop

categories (500 control points in total), in order to prevent statistical bias in favor of crops that are common in the area. The same control points were used to evaluate both models.

- **Radiometric cross-referencing:** The control point layer was overlaid on each of the classified rasters, and the predicted values of the models were extracted into the attribute table of the points using the Update Accuracy Assessment Points tool.
- **Confusion Matrix:** The final accuracy analysis was quantified using the Compute Confusion Matrix tool. Two separate matrices were produced (one for KNN and one for RF), which provided a quantitative statistical comparison between the verified classification and the model prediction. From the matrices, the standard accuracy metrics were derived: Overall Accuracy, Producer's Accuracy, and User's Accuracy. The statistical results were reported to a precision of two significant digits, which made it possible to determine the optimal algorithm for classification in this area.

3. Results

This chapter presents the results of the spatial classification of agricultural crops in the Yoav Regional Council, as obtained from the two machine learning models: Random Forest (RF) and K-Nearest Neighbors (KNN). The Accuracy Assessment was based on an independent sample of 494 valid control points.

3.1 Overall Performance Comparison

An examination of the summary accuracy metrics indicates a gap in the performance of the two models, with the RF model demonstrating superior classification capability over the KNN model across all metrics examined. Table 1 presents the Overall Accuracy and the Kappa Coefficient values for each of the algorithms.

Classification model	Overall Accuracy	Kappa Coefficient
Random Forest (RF)	51.20%	0.39
K- Nearest Neighbors	46.10%	0.327

Table 1: Summary accuracy metrics for the comparison of the classification models

These data indicate that the use of a decision-tree ensemble model (RF) enables better handling of the spectral-temporal complexity of the agricultural area, compared with a distance-based model (KNN). This performance gap effectively corroborates the established observations in remote sensing applications, according to which there is strong evidence that Ensemble-based machine learning methods are more consistent and efficient than standard single classifiers (Maxwell et al., 2018).

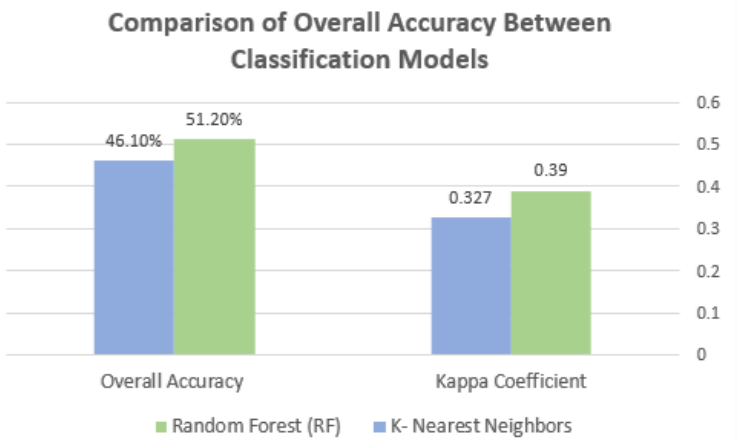


Figure 1: Overall performance comparison between the RF model and the KNN model.

3.2 Confusion Matrices Analysis

In order to conduct a focused examination of classification quality at the level of the individual crop categories, confusion matrices were produced for both models. The analysis of the matrices makes it possible to identify the accuracy levels specific to each crop type (Producer's and User's Accuracy), as well as to detect systematic Misclassification patterns between crops with similar spectral signatures.

Confusion_Matrix_RF

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	OBJECTID *	ClassValue	Avocado	Olive	Wheat	Corn	Clover	Total	U_Accuracy	Kappa
1	1	C_1	71	18	11	8	10	118	0.601695	0
2	2	C_2	15	71	17	12	18	133	0.533835	0
3	3	C_3	7	4	26	23	21	81	0.320988	0
4	4	C_4	2	4	14	49	14	83	0.590361	0
5	5	C_5	3	2	32	6	36	79	0.455696	0
6	6	Total	98	99	100	98	99	494	0	0
7	7	P_Accuracy	0.72449	0.717172	0.26	0.5	0.363636	0	0.512146	0
8	8	Kappa	0	0	0	0	0	0	0	0.390248

Table 2: Confusion Matrix – Random Forest model

KNN_Matrix_Final.csv

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	ClassValue	Avocado	Olive	Wheat	Corn	Clover	Total	U_Accuracy	Kappa
1	C_1	58	22	9	7	8	104	0.557	0
2	C_2	17	52	11	12	7	99	0.525	0
3	C_3	9	6	21	17	20	73	0.287	0
4	C_4	4	9	24	52	19	108	0.481	0
5	C_5	10	10	35	10	45	110	0.409	0
6	Total	98	99	100	98	99	494	0	0
7	P_Accuracy	0.591	0.525	0.21	0.531	0.455	0	0.461	0
8	Kappa	0	0	0	0	0	0	0	0.327

Table 3: Confusion Matrix – K-Nearest Neighbors model

A detailed examination of the results of the **KNN** algorithm reveals considerable variability in the ability to identify different crops:

- **Highest accuracy level:** This was obtained for the **avocado** crop, which achieved a Producer's Accuracy of **59.1%**. This indicates that this crop is characterized by a relatively distinctive spectral signature, which the model was able to identify with the highest success rate. A similar trend was also observed for the **olive** crop (**52.5%**), which attests to the relative ability of the model to identify tree orchards.
- **Lowest accuracy level and error patterns:** By contrast, the model struggled considerably to identify seasonal field crops, and in particular **wheat**, which attained a low Producer's Accuracy of only **21.0%**. Analysis of the confusion matrix reveals that most of the errors in this category resulted from misclassification (confusion) with **clover** (35 misclassified points) and **corn** (24 misclassified points). This result suggests a high degree of

similarity in the phenology and growth rate of the field crop group, which makes it difficult for the KNN algorithm to produce an accurate spatial separation between them.

An examination of the results of the **Random Forest** algorithm indicates a similar trend in classification patterns, but with a marked improvement in overall performance, and in particular in the ability to separate tree orchards:

- **The leap in orchard accuracy:** The **avocado** and **olive** crops were identified with the highest level of accuracy, with Producer's Accuracy values of **72.4%** and **71.7%**, respectively. This represents a dramatic improvement over the KNN model, a finding that highlights the clear advantage of tree-based algorithms in handling the multi-temporal spectral signatures of evergreen orchards in the agricultural landscape.
- **The field crop challenge:** Despite the superiority of the RF model, the field crop group remained challenging. The **wheat** crop exhibited the lowest level of accuracy (**26.0%**). As with KNN, most of the "confusion" occurred with **clover** (32 misclassified points out of 100). This finding reinforces the conclusion that the identification limitation stems from an inherent phenological-spectral similarity between these crops during their growth stages, a difficulty that even an advanced ensemble algorithm struggles to fully resolve.

3.3 Visual Representation of Spatial Classification

Beyond the statistical metrics, a spatial examination of the classification maps provides important insights into how the algorithms perform on the ground.

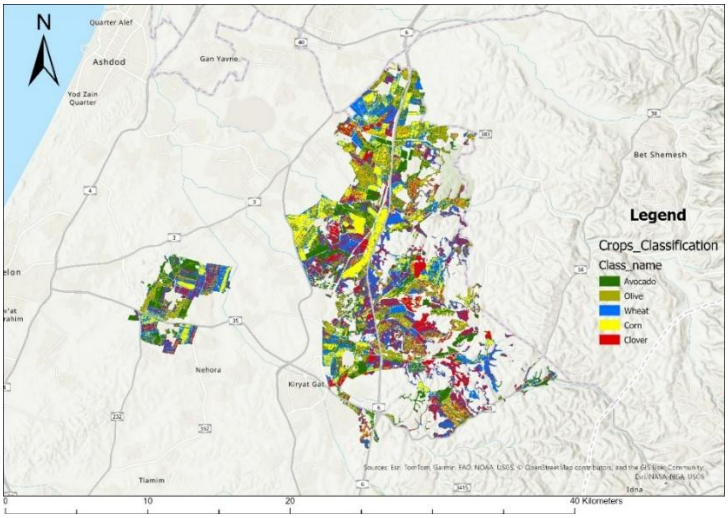


Figure 2: Agricultural crop classification map as produced by the Random Forest model.

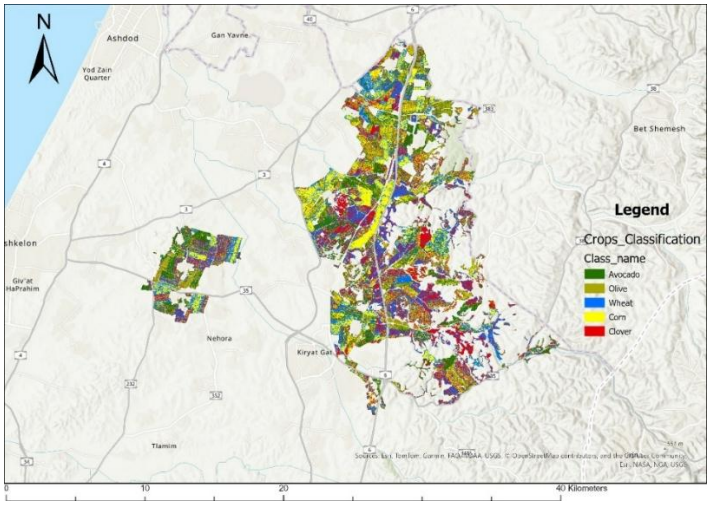


Figure 3: Agricultural crop classification map as produced by the K-Nearest Neighbors model.

A visual comparison of the classification maps demonstrates the spatial differences between the models. Whereas the KNN model tends to produce a "speckled" and noisier output at the individual pixel level, the Random Forest model succeeds in producing more homogeneous and contiguous parcels. This homogeneity more faithfully reflects the physical reality of managed agricultural parcels and reinforces the preference for this model as a mapping tool in the given area.

In order to illustrate the spatial differences between the outputs of the models, a focused local examination was carried out in a land unit around Kibbutz Sde Yoav. A visual comparison of this area clearly demonstrates the superiority of the Random Forest model in representing the reality on the ground.

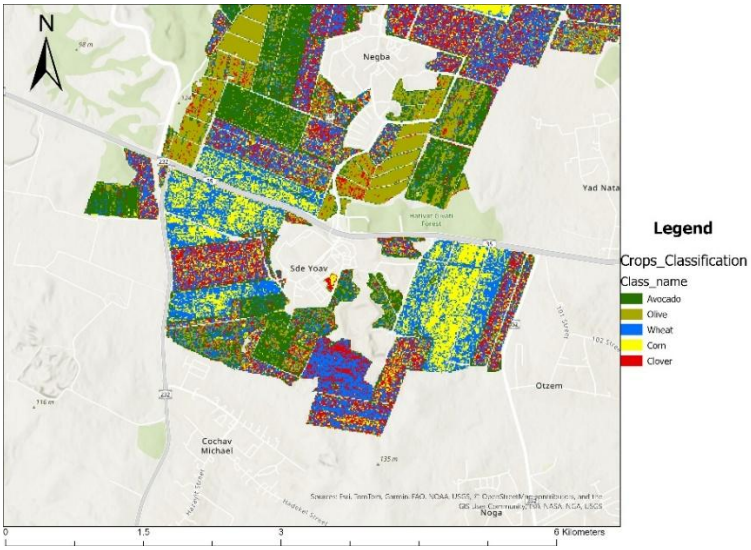


Figure 4: Agricultural crop classification map as produced by the K-Nearest Neighbors model, focusing on Kibbutz Sde Yoav and Kibbutz Negba.

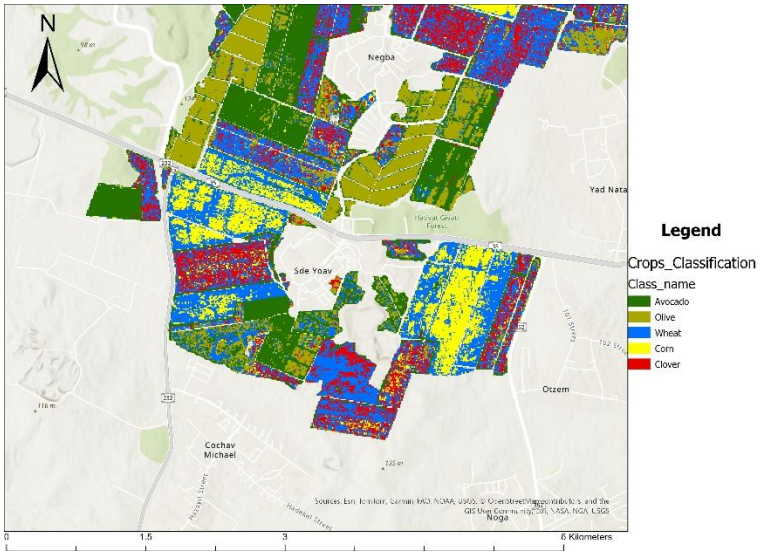


Figure 5: Agricultural crop classification map as produced by the Random Forest model, focusing on Kibbutz Sde Yoav and Kibbutz Negba.

In the prediction map of the KNN algorithm (Figure 4), a very high level of noise is evident at the individual pixel level. It can be seen that within the boundaries of a single agricultural parcel, pixels classified as a variety of different crops are intermixed (for example, isolated wheat and corn pixels scattered within clover fields or orchards). This fragmentation contradicts the agronomic logic of managing uniform commercial parcels. By contrast, in the prediction map of the RF model (Figure 5), the agricultural parcels emerge as homogeneous, contiguous spaces, delineated with higher quality than the KNN model was able to predict. The ability of the RF algorithm to aggregate multiple decision trees enabled it to filter out local spectral “background noise” and to produce a contiguous classification that faithfully reflects the actual agricultural patterns on the ground.

4. Discussion and Conclusions

This study examined the ability to automatically classify agricultural crops in the Yoav Regional Council, based on free remote sensing data from the Sentinel-2 satellite and the use of machine learning algorithms. The results of the study clearly indicate the superiority of the Random Forest (RF) model over the K-Nearest Neighbors (KNN) model, both in the bare statistical metrics (Overall Accuracy and Kappa) and in the spatial-visual analysis of the classification outputs.

The Methodological Advantage of a Tree-Based Algorithm

The performance gap between the two models stems from their decision-making architecture. Whereas the KNN algorithm classifies pixels on the basis of proximity in the feature space (Euclidean distance), it struggles to cope with "noise" in the NDVI time series and tends to produce a fragmented classification (a pixelated image). In contrast, the RF model, as an ensemble algorithm that examines subsets of variables through multiple decision trees, exhibits greater robustness to spectral noise (Belgiu & Drăguţ, 2016). This capability enabled RF to identify the variables most critical for separation and to create classification maps with a spatial homogeneity that faithfully reflects the agronomic reality of contiguous parcels. This built-in mechanism for assessing Variable Importance constitutes a prominent analytical advantage of the algorithm, since it enables researchers to quantify precisely which feature, or which month within the time series, contributes most significantly to the spectral separability between crop types (Belgiu & Drăguţ, 2016).

This trend is supported by the professional literature, which indicates that distance-based models such as KNN are particularly sensitive to the "curse of dimensionality" (The Hughes Phenomenon) and to data abundance, whereas ensemble-based algorithms such as RF successfully handle complex feature spaces without a considerable loss of accuracy (Maxwell et al., 2018).

The Effect of Phenology on Spectral Separability

The analysis of the confusion matrices reveals a compelling picture regarding the relationship between crop type and its identifiability. Both models, and RF in particular, performed very well in identifying orchards (avocado and olive, over 71% accuracy for RF). This success is attributed to the fact that these are evergreen trees, which produce a stable and relatively high spectral signature throughout most months of the year, making it easier for the algorithm to isolate them from other crops.

On the other hand, the main challenge that remained not fully addressed is the identification of seasonal field crops, foremost among them wheat and clover. The high error rate between these two crops stems from a near-complete phenological similarity: both are winter/spring crops that are sown, green up, and senesce within parallel time windows. As a result, their NDVI curves over the growing season are almost entirely identical. This finding demonstrates that under conditions of tight phenological overlap, spectral and temporal resolution

alone is insufficient, and that it may be necessary in the future to integrate radar (SAR) data, which are sensitive to the physical structure of the plant and not only to its greenness.

Summary and Practical Implications

The study demonstrates that combining free Sentinel-2 data with the Random Forest algorithm constitutes a practical, accessible, and cost-effective tool for agricultural monitoring, and exhibits advanced capabilities in handling a complex agricultural landscape. Beyond the value of the mapping itself, the production of reliable information layers on crop distribution constitutes an essential infrastructure for decision-making in rural areas and for a more sustainable management of land and agricultural resources. At the same time, the findings of the study and the challenges that emerged from it, such as the difficulty in separating field crops with phenological overlap, highlight that reliance on spectral resolution at the individual pixel level is not sufficient on its own. This insight paves the way for advanced follow-up studies that will focus on the transition from pixel-based classification to object-based classification (OBIA), as well as on the integration of complementary structural data. Future development of these methodologies will make it possible to maximize the accuracy of crop maps, all while strictly adhering to the principle of using freely available data sources.

Future Research Directions: From Pixel-Based to Object-Based Classification (OBIA)

A potential and significant development for follow-up studies is upgrading the methodology from classification at the individual pixel level (Pixel-based classification) to object-based classification (OBIA). Whereas the current models examine each land cell separately, an object-based model would first identify the boundaries of agricultural parcels as complete polygons. The research justification for this approach rests on the physical practice in the field: farmers sow and cultivate land units as contiguous and uniform blocks, not in random segments. In addition, pixel-level classification is usually susceptible to “Salt-and-Pepper” noise, whereas combining RF with the OBIA approach aggregates pixels into homogeneous spatial units that more accurately reflect the structure of the agricultural landscape (Belgiu & Drăguţ, 2016).

This agronomic understanding makes an object-based algorithm particularly relevant and accurate for agriculture, since it adapts itself to the spatial reality.

On top of the contiguous polygons, a “Majority Voting” mechanism can be applied, which analyzes the distribution of pixels within the parcel. Thus, for example, a parcel in which 70% of the pixels were identified as wheat and 30% as clover would be classified in its entirety as a homogeneous wheat parcel, while effectively neutralizing local spectral “noise.” In addition, it is recommended to incorporate into future models a quality control mechanism based on a “Confidence Threshold.” Under this mechanism, a parcel that does not exceed a predefined dominance threshold of pixels of a particular type would not be misclassified, but would instead be labeled under an “Unclassified” category. Implementing a strict threshold value would significantly reduce the rate of “False Positives” and ensure that the final crop maps are reliable enough for regulatory and environmental planning purposes.

However, from a critical research perspective, it is important to note that the theoretical advantages of OBIA may be limited when Sentinel-2 imagery is used. Pioneering studies comparing the approaches have shown that at the given spatial resolution (10 meters), object-based classification does not necessarily guarantee a significant improvement over pixel-based classification with the RF algorithm, particularly in agricultural environments characterized by small or fragmented parcels in which it is difficult to delineate homogeneous objects. This finding reinforces the conclusion that RF-based pixel classification already constitutes a strong and reliable starting point (Immitzer et al., 2016).

The Challenge of Classifying Evergreen Orchards and the Integration of Free Structural Data and Texture Analysis (SAR)

Although the models in this study, and in particular Random Forest, performed very well in separating avocado and olive trees from the remaining field crops, it is important to note that this success relies to a large extent on the clear difference between the spectral signature of a fruit tree orchard and that of an ordinary seasonal crop. However, a significant challenge in follow-up studies will be the ability to distinguish between avocado trees and other evergreen crops with similar spectral signatures and foliage, such as citrus groves. Since these crops share nearly identical chlorophyll density levels and phenology, reliance on spectral resolution alone, such as the NDVI index, may prove insufficient and lead to misclassification.

In order to overcome this limitation while adhering to the methodological principle of using accessible and free remote sensing data, it is recommended to incorporate into future models variables based on the physical structure of the parcel. Each orchard type is characterized by a distinctive "Spatial Roughness" and landscape configuration. This structure can be quantified at no cost through two complementary approaches. First, applying image texture analysis algorithms (Texture Analysis), such as the GLCM (Gray Level Co-occurrence Matrix) metric, directly to the existing Sentinel-2 imagery. This statistical method quantifies the spatial relationships of brightness values between each pixel and its surroundings within a defined window (Kernel), thereby translating the physical structure of the vegetation into numerical metrics. Recent studies corroborate this direction, demonstrating that the incorporation of GLCM-based texture features effectively compensates for spectral overlap between crops. Moreover, adjusting the window size to the characteristics of the specific crop improves classification accuracy and significantly reduces local classification noise (Kwak & Park, 2019).

Second, adding free radar (SAR) data from the Sentinel-1 satellite. Radar waves are sensitive to the geometric structure and water content of vegetation and can provide the learning model with a physical-structural dimension in addition to the spectral dimension. Recent studies combining Sentinel-1 time series with optical data corroborate this potential and demonstrate how tracking changes in Volume Scattering mechanisms, that is, the manner in which radar waves are scattered from the three-dimensional structure of the plant, such as the network of stems and leaves, over the course of the growing season makes it possible to accurately characterize the physiological structure of the vegetation (Xu et al., 2018). In particular, analysis of the VV/VH

polarization ratio provides unique structural information that is not present in optical indices, thereby offering an effective alternative solution for separating crops with completely identical spectral signatures or NDVI profiles (Veloso et al., 2017).

The addition of these structural variables is expected to considerably increase the separability between different types of orchards, without increasing the costs of continuous monitoring.

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